РОССИЙСКИЙ УНИВЕРСИТЕТ ДРУЖБЫ НАРОДОВ

Факультет физико-математических и естественных наук

ОТЧЁТ

ПО ЛАБОРАТОРНОЙ РАБОТЕ №7

дисциплина: Администрирование сетевых подсистем

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Группа: НФИбд-02-23

МОСКВА

2025 г.

Цель работы:

Целью данной работы является получение навыков настройки межсетевого экрана в Linux в части переадресации портов и настройки Masquerading.

Выполнение работы:

root@server:/etc/firewalld/services

```
[vagrant@server ~]$ sudo -i
[root@server ~]# cp /usr/lib/firewalld/services/ssh.xml /etc/firewalld/services/
ssh-custom.xml
[root@server ~]# cd /etc/firewalld/services/
[root@server services]# cat /etc/firewalld/services/ssh-custom.xml
<?xml version="1.0" encoding="utf-8"?>
<service>
  <short>SSH</short>
  <description>Secure Shell (SSH) is a protocol for logging into and executing c
ommands on remote machines. It provides secure encrypted communications. If you
plan on accessing your machine remotely via SSH over a firewalled interface, ena
ble this option. You need the openssh-server package installed for this option t
o be useful.</description>
  <port protocol="tcp" port="22"/>
</service>
[root@server services]# gedit /etc/firewalld/services/ssh-custom.xml

(gedit:19372): dconf-WARNING **: 13:04:50.569: failed to commit changes to dconf
: Failed to execute child process “dbus-launch” (No such file or directory)

(gedit:19372): dconf-WARNING **: 13:04:50.586: failed to commit changes to dconf
: Failed to execute child process “dbus-launch” (No such file or directory)

(gedit:19372): dconf-WARNING **: 13:04:50.868: failed to commit changes to dconf
```

Open ▾ *ssh-custom.xml /etc/firewalld/services Save

```xml
<?xml version="1.0" encoding="utf-8"?>
<service>
  <short>SSH</short>
  <description>This a modified file</description>
  <port protocol="tcp" port="2022"/>
</service>
```

root@server:/etc/firewalld/services

```
t-encoding not supported

** (gedit:19372): WARNING **: 13:08:59.634: Set document metadata failed: Setting attribute metadata::gedi
t-position not supported

(gedit:19372): dconf-WARNING **: 13:08:59.642: failed to commit changes to dconf: Failed to execute child
process “dbus-launch” (No such file or directory)
[root@server services]# firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp amqps apcupsd audit ausweisa
pp2 bacula bacula-client bareos-director bareos-filedaemon bareos-storage bb bgp bitcoin bitcoin-rpc bitco
in-testnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-exporter ceph-mon cfengine checkmk-agent cockpit c
ollectd condor-collector cratedb ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns d
ns-over-tls docker-registry docker-swarm dropbox-lansync elasticsearch etcd-client etcd-server finger fore
man foreman-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera gangli
a-client ganglia-master git gpsd grafana gre high-availability http http3 https ident imap imaps ipfs ipp
ipp-client ipsec irc ircs iscsi-target isns jenkins kadmin kdeconnect kerberos kibana klogin kpasswd kprop
 kshell kube-api kube-apiserver kube-control-plane kube-control-plane-secure kube-controller-manager kube-
controller-manager-secure kube-nodeport-services kube-scheduler kube-scheduler-secure kube-worker kubelet
kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-network llmnr llmnr-client llmnr-
tcp llmnr-udp managesieve matrix mdns memcache minidlna mongodb mosh mountd mqtt mqtt-tls ms-wbt mssql mur
mur mysql nbd nebula netbios-ns netdata-dashboard nfs nfs3 nmea-0183 nrpe ntp nut openvpn ovirt-imageio ov
irt-storageconsole ovirt-vmconsole plex pmcd pmproxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy prom
etheus prometheus-node-exporter proxy-dhcp ps2link ps3netsrv ptp pulseaudio puppetmaster quassel radius rd
p redis redis-sentinel rpc-bind rquotad rsh rsyncd rtsp salt-master samba samba-client samba-dc sane sip s
ips slp smtp smtp-submission smtps snmp snmptls snmptls-trap snmptrap spideroak-lansync spotify-sync squid
 ssdp ssh steam-streaming svdrp svn syncthing syncthing-gui syncthing-relay synergy syslog syslog-tls teln
et tentacle tftp tile38 tinc tor-socks transmission-client upnp-client vdsm vnc-server warpinator wbem-htt
p wbem-https wireguard ws-discovery ws-discovery-client ws-discovery-tcp ws-discovery-udp wsman wsmans xdm
cp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabbix-server zerotier
[root@server services]#
```

```
root@server:/etc/firewalld/services
cp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabbix-server zerotier
[root@server services]# irewall-cmd --reload
bash: irewall-cmd: command not found...
[root@server services]# firewall-cmd --reload
success
[root@server services]# firewall-cmd --get-services
RH-Satellite-6 RH-Satellite-6-capsule afp amanda-client amanda-k5-client amqp amqps apcupsd audit ausweisa
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ollectd condor-collector cratedb ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns d
ns-over-tls docker-registry docker-swarm dropbox-lansync elasticsearch etcd-client etcd-server finger fore
man foreman-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera gangli
a-client ganglia-master git gpsd grafana gre high-availability http http3 https ident imap imaps ipfs ipp
ipp-client ipsec irc ircs iscsi-target isns jenkins kadmin kdeconnect kerberos kibana klogin kpasswd kprop
 kshell kube-api kube-apiserver kube-control-plane kube-control-plane-secure kube-controller-manager kube-
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etheus prometheus-node-exporter proxy-dhcp ps2link ps3netsrv ptp pulseaudio puppetmaster quassel radius rd
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 ssdp ssh ssh-custom steam-streaming svdrp svn syncthing syncthing-gui syncthing-relay synergy syslog sysl
og-tls telnet tentacle tftp tile38 tinc tor-socks transmission-client upnp-client vdsm vnc-server warpinat
or wbem-http wbem-https wireguard ws-discovery ws-discovery-client ws-discovery-tcp ws-discovery-udp wsman
 wsmans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabbix-server zerotier
[root@server services]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client http https imap imaps pop3 pop3s smtp ssh
[root@server services]#
```

```
root@server:/etc/firewalld/services
cp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabbix-server zerotier
[root@server services]# irewall-cmd --reload
bash: irewall-cmd: command not found...
[root@server services]# firewall-cmd --reload
success
[root@server services]# firewall-cmd --get-services
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in-testnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-exporter ceph-mon cfengine checkmk-agent cockpit c
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man foreman-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera gangli
a-client ganglia-master git gpsd grafana gre high-availability http http3 https ident imap imaps ipfs ipp
ipp-client ipsec irc ircs iscsi-target isns jenkins kadmin kdeconnect kerberos kibana klogin kpasswd kprop
 kshell kube-api kube-apiserver kube-control-plane kube-control-plane-secure kube-controller-manager kube-
controller-manager-secure kube-nodeport-services kube-scheduler kube-scheduler-secure kube-worker kubelet
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 ssdp ssh ssh-custom steam-streaming svdrp svn syncthing syncthing-gui syncthing-relay synergy syslog sysl
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 wsmans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabbix-server zerotier
[root@server services]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client http https imap imaps pop3 pop3s smtp ssh
[root@server services]#
```
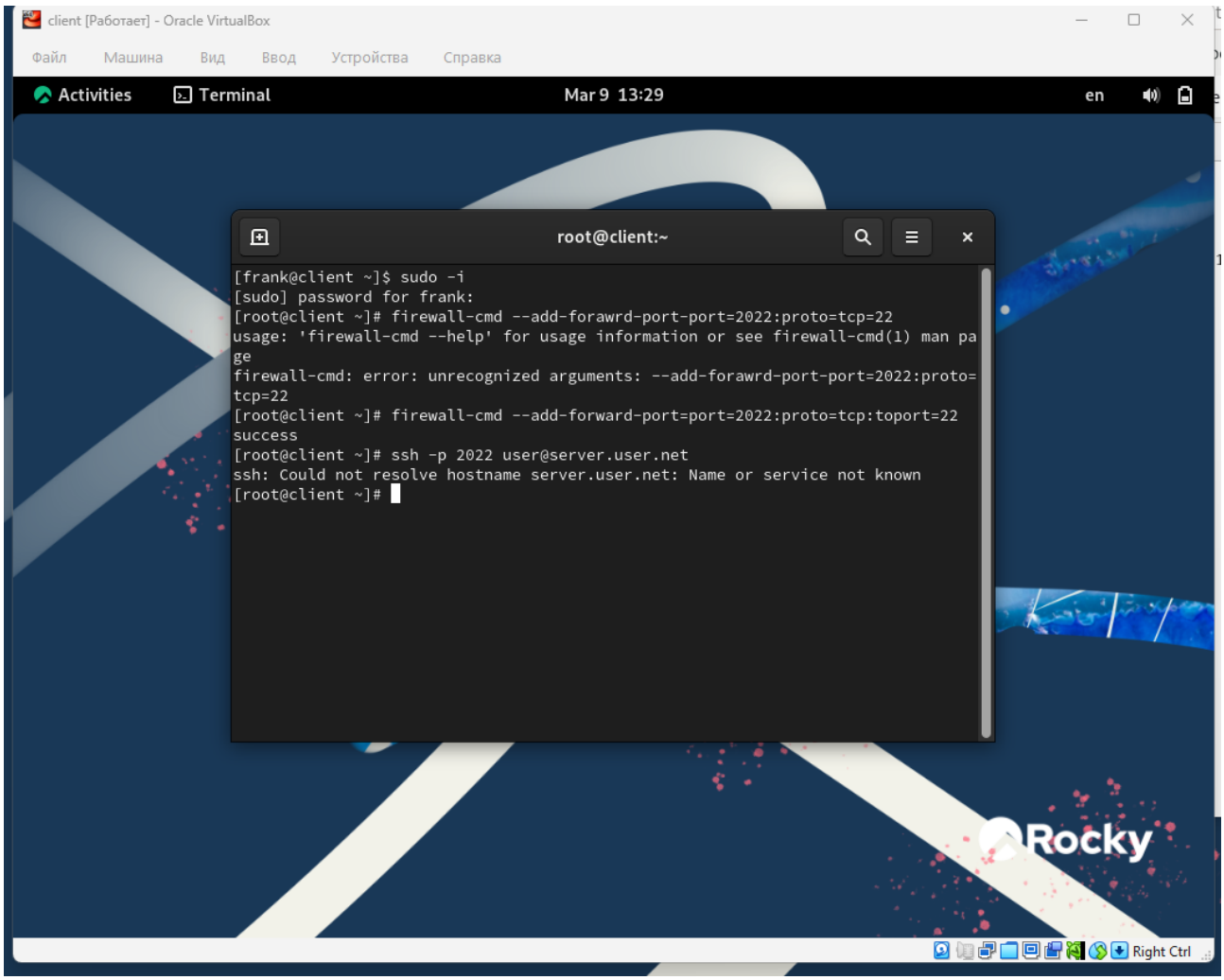

```
 wsmans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-agent zabbix-server zerotier
[root@server services]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client http https imap imaps pop3 pop3s smtp ssh
[root@server services]# firewall-cmd --add-service=ssh-custom
success
[root@server services]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client http https imap imaps pop3 pop3s smtp ssh ssh-custom
[root@server services]# firewall-cmd --add-service=ssh-custom --permanent
success
[root@server services]# firewall-cmd --reload
success
[root@server services]#
```


client [Работает] - Oracle VirtualBox
Файл Машина Вид Ввод Устройства Справка
Activities
Terminal
Mar 9 13:29
en
root@client:~
[frank@client ~]$ sudo -i
[sudo] password for frank:
[root@client ~]# firewall-cmd --add-forawrd-port-port=2022:proto=tcp=22
usage: 'firewall-cmd --help' for usage information or see firewall-cmd(1) man page
firewall-cmd: error: unrecognized arguments: --add-forawrd-port-port=2022:proto=tcp=22
[root@client ~]# firewall-cmd --add-forward-port=port=2022:proto=tcp:toport=22
success
[root@client ~]# ssh -p 2022 user@server.user.net
ssh: Could not resolve hostname server.user.net: Name or service not known
[root@client ~]#
Rocky
Right Ctrl

```
root@server:/etc/firewalld/services
[root@server services]# firewall-cmd --add-forward-port=port=2022:proto=tcp:toport=22
success
[root@server services]# ssh -p 2022 user@server.user.net
ssh: connect to host server.user.net port 2022: Connection refused
[root@server services]# sysctl -a | grep forward
net.ipv4.conf.all.bc_forwarding = 0
net.ipv4.conf.all.forwarding = 0
net.ipv4.conf.all.mc_forwarding = 0
net.ipv4.conf.default.bc_forwarding = 0
net.ipv4.conf.default.forwarding = 0
net.ipv4.conf.default.mc_forwarding = 0
net.ipv4.conf.eth0.bc_forwarding = 0
net.ipv4.conf.eth0.forwarding = 0
net.ipv4.conf.eth0.mc_forwarding = 0
net.ipv4.conf.eth1.bc_forwarding = 0
net.ipv4.conf.eth1.forwarding = 0
net.ipv4.conf.eth1.mc_forwarding = 0
net.ipv4.conf.lo.bc_forwarding = 0
net.ipv4.conf.lo.forwarding = 0
net.ipv4.conf.lo.mc_forwarding = 0
net.ipv4.ip_forward = 0
net.ipv4.ip_forward_update_priority = 1
net.ipv4.ip_forward_use_pmtu = 0
net.ipv6.conf.all.forwarding = 0
net.ipv6.conf.all.mc_forwarding = 0
net.ipv6.conf.default.forwarding = 0
net.ipv6.conf.default.mc_forwarding = 0
net.ipv6.conf.eth0.forwarding = 0
net.ipv6.conf.eth0.mc_forwarding = 0
net.ipv6.conf.eth1.forwarding = 0
```

```
net.ipv4.ip_forward_use_pmtu = 0
net.ipv6.conf.all.forwarding = 0
net.ipv6.conf.all.mc_forwarding = 0
net.ipv6.conf.default.forwarding = 0
net.ipv6.conf.default.mc_forwarding = 0
net.ipv6.conf.eth0.forwarding = 0
net.ipv6.conf.eth0.mc_forwarding = 0
net.ipv6.conf.eth1.forwarding = 0
net.ipv6.conf.eth1.mc_forwarding = 0
net.ipv6.conf.lo.forwarding = 0
net.ipv6.conf.lo.mc_forwarding = 0
[root@server services]# echo "net.ipv4.ip_forward = 1" > /etc/sysctl.d/90-forward.conf
[root@server services]# sysctl -p /etc/sysctl.d/90-forward.conf
net.ipv4.ip_forward = 1
[root@server services]# firewall-cmd --zone=public --add-masquerade --permanent
success
[root@server services]# firewall-cmd --reload
success
[root@server services]#
```

Open ▾ | firewall.sh /vagrant/provision/server | Save

```
1 #!/bin/bash
2 echo "Provisioning script $0"
3 echo "Copy configuration files"
4 cp -R /vagrant/provision/server/firewall/etc/* /etc
5 echo "Configure masquerading"
6 firewall-cmd --add-service=ssh-custom --permanent
7 firewall-cmd --add-forward-port=port=2022:proto=tcp:toport=22 --permanent
8 firewall-cmd --zone=public --add-masquerade --permanent
9 firewall-cmd --reload
10 restorecon -vR /etc
```

sh ▾ Tab Width: 8 ▾ Ln 10, Col 20 ▾ INS

Open Save

firewall.sh
/vagrant/provision/server

```
#!/bin/bash
echo "Provisioning script $0"
echo "Copy configuration files"
cp -R /vagrant/provision/server/firewall/etc/* /etc
echo "Configure masquerading"
firewall-cmd --add-service=ssh-custom --permanent
firewall-cmd --add-forward-port=port=2022:proto=tcp:toport=22 --permanent
firewall-cmd --zone=public --add-masquerade --permanent
firewall-cmd --reload
restorecon -vR /etc
```

sh Tab Width: 8 Ln 10, Col 20 INS

Вывод:

В ходе выполнения лабораторной работы были получены навыки настройки межсетевого экрана в Linux в части переадресации портов и настройки Masquerading.

Ответы на контрольные вопросы:

1. Где хранятся пользовательские файлы firewalld? - **В firewalld пользовательские файлы хранятся в директории /etc/firewalld/.**
2. Какую строку надо включить в пользовательский файл службы, чтобы указать порт TCP 2022? - **Для указания порта TCP 2022 в пользовательском файле службы, вы можете добавить строку в секцию port следующим образом:**

   <port protocol="tcp" port="2022"/>

3. Какая команда позволяет вам перечислить все службы, доступные в настоящее время на вашем сервере? - **Чтобы перечислить все службы, доступные в настоящее время на сервере с использованием firewalld, используйте команду:**

   firewall-cmd --get-services

4. В чем разница между трансляцией сетевых адресов (NAT) и маскарадингом (masquerading)? - **Разница между трансляцией сетевых адресов (NAT) и маскарадингом (masquerading)**

заключается в том, что в случае NAT исходный IP-адрес пакета заменяется на IP-адрес маршрутизатора, а в случае маскарадинга используется IP-адрес интерфейса маршрутизатора.

5. Какая команда разрешает входящий трафик на порт 4404 и перенаправляет его в службу ssh по IP-адресу 10.0.0.10? - **Для разрешения входящего трафика на порт 4404 и перенаправления его на службу SSH по IP-адресу 10.0.0.10, вы можете использовать команды:**

   firewall-cmd --zone=public --add-port=4404/tcp --permanent

   firewall-cmd --zone=public --add-forward-port=port=4404:proto=tcp:toport=22:toaddr=10.0.0.10 --permanent

   firewall-cmd --reload

6. Какая команда используется для включения маскарадинга IP-пакетов для всех пакетов, выходящих в зону public? - **Для включения маскарадинга IP-пакетов для всех пакетов, выходящих в зону public, используйте следующую команду:**

   firewall-cmd --zone=public --add-masquerade --permanent

   firewall-cmd --reload